AuthX - Break the Login: Security Audit Report

A Flask web application built in two versions to demonstrate common authentication vulnerabilities and their fixes.

Team Members

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- Assisted by Copilot (Claude Sonnet 4.6) for polishing documentation and increasing coverage of security fixes.

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1. Introduction

1.1 Application Description

AuthX is a fictional internal portal used by employees to manage security tickets. The application supports two roles:

- **ANALYST** - can create, view, edit, and delete their own tickets
- **MANAGER** - can view and manage all tickets and access the audit log

The project covers the full authentication layer (login, registration, session management, password reset) as well as ticket CRUD operations, demonstrating both authentication failures and broken access control vulnerabilities. It was developed in two versions:

Version	Purpose
`v1-vulnerable`	Intentionally insecure baseline for PoC demonstrations
`v2-secure`	Hardened implementation with all fixes applied

1.2 Architecture

```
v1-vulnerable/          v2-secure/
├── app.py              ├── app.py
├── models.py           ├── models.py
├── requirements.txt    ├── requirements.txt
├── environment.yml     ├── environment.yml
└── templates/          └── templates/
    ├── base.html           ├── base.html
    ├── login.html          ├── login.html
    ├── register.html       ├── register.html
    ├── dashboard.html      ├── dashboard.html
    ├── forgot.html         ├── forgot.html
    ├── reset.html          ├── reset.html
    ├── tickets.html        ├── tickets.html
    └── audit.html          └── audit.html

poc/
├── poc_51_52_weak_password_storage.py
├── poc_53_brute_force.py
├── poc_54_user_enumeration.py
├── poc_55_insecure_session.py
└── poc_56_reset_token.py
```

Stack: Python 3.11, Flask 3.0, SQLite 3
Database tables: `users`, `tickets`, `audit_logs`

1.3 Database Schema

```
users (id, email, password, role, created_at, locked)
tickets (id, title, description, severity, status, owner_id, created_at,
updated_at)
audit_logs (id, user_id, action, resource, resource_id, ip_address, timestamp)
```

---

2. Environment Setup

2.1 Prerequisites

- Miniconda (or Anaconda)

2.2 Installation

Each version ships an `environment.yml` that pins all dependencies.

```
# v1 - vulnerable
conda env create -f v1-vulnerable/environment.yml
conda activate dass-v1

# v2 - secure
conda env create -f v2-secure/environment.yml
conda activate dass-v2
```

2.3 Running v1 (vulnerable)

```
cd v1-vulnerable
conda run -n dass-v1 --no-capture-output python app.py
```

Open `http://127.0.0.1:5000`

2.4 Running v2 (secure)

```
cd v2-secure
conda run -n dass-v2 --no-capture-output python app.py
```

Open `http://127.0.0.1:5001`

2.5 Default Accounts

Neither version seeds the database. Register accounts manually via `/register`. Role defaults to `ANALYST`.

To promote an account to **MANAGER**, run the following from the version's directory (where `authx.db` lives):

```
python -c "
import sqlite3
conn = sqlite3.connect('authx.db')
conn.execute(\"UPDATE users SET role='MANAGER' WHERE email='your@email.com'\")
conn.commit()
"
```

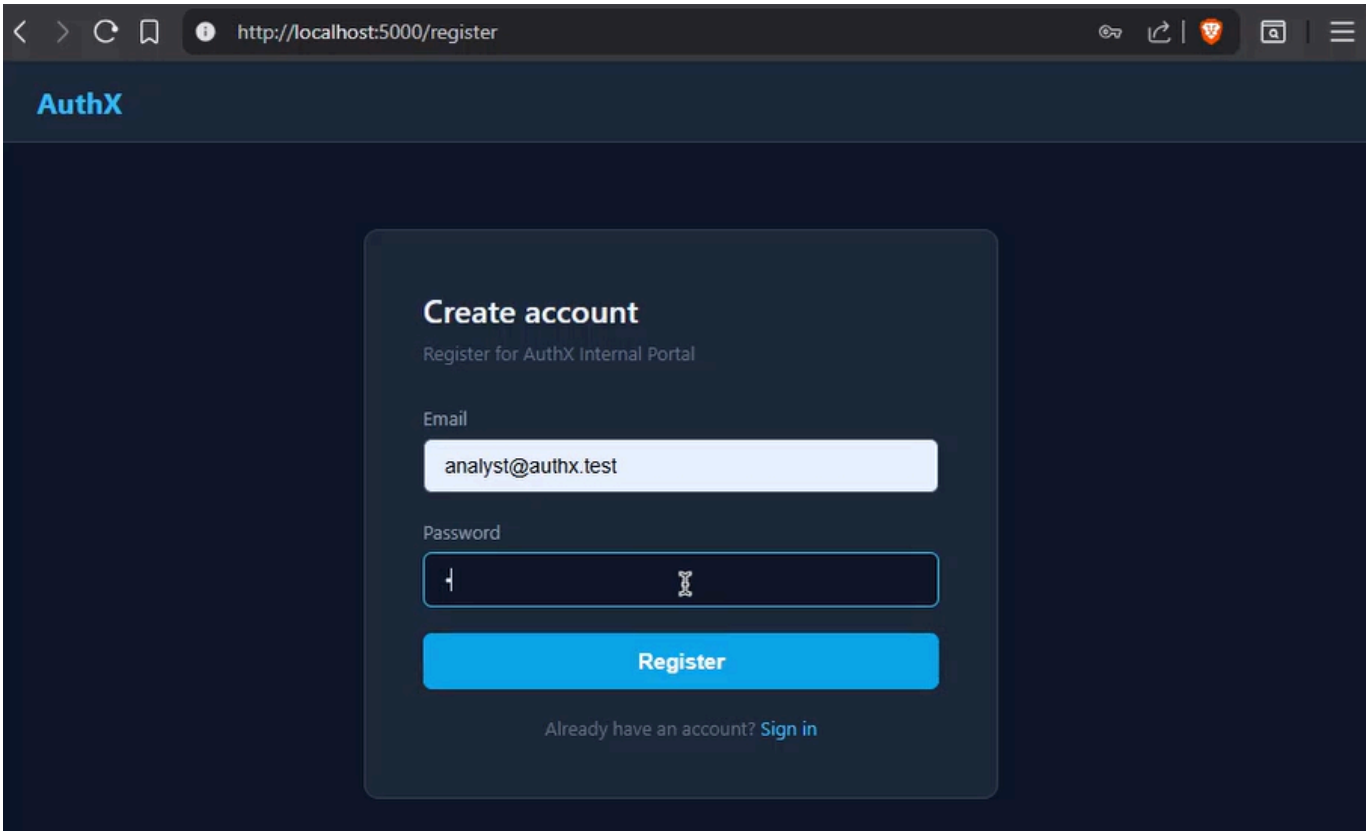
---

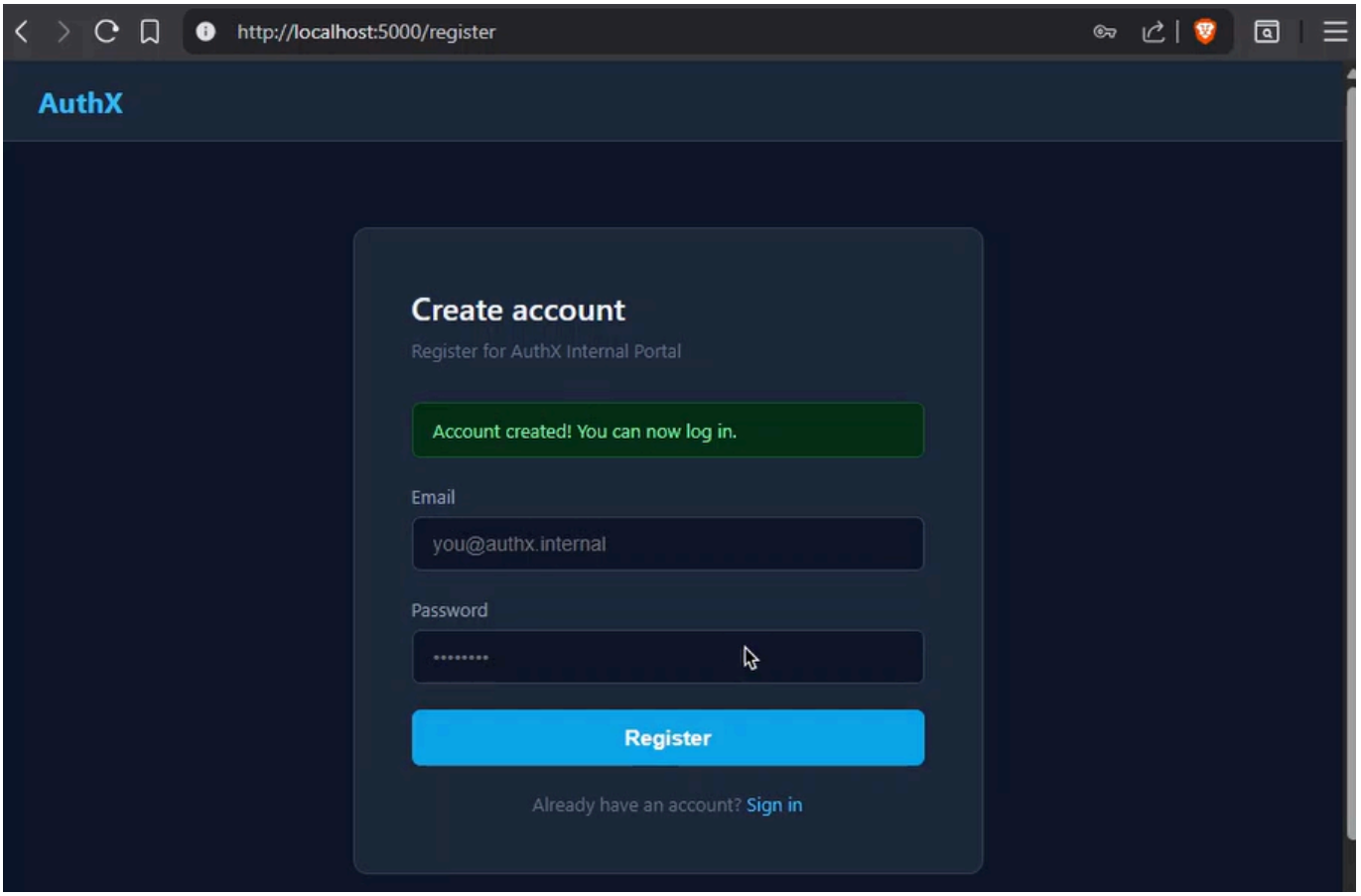
3. MVP Implementation

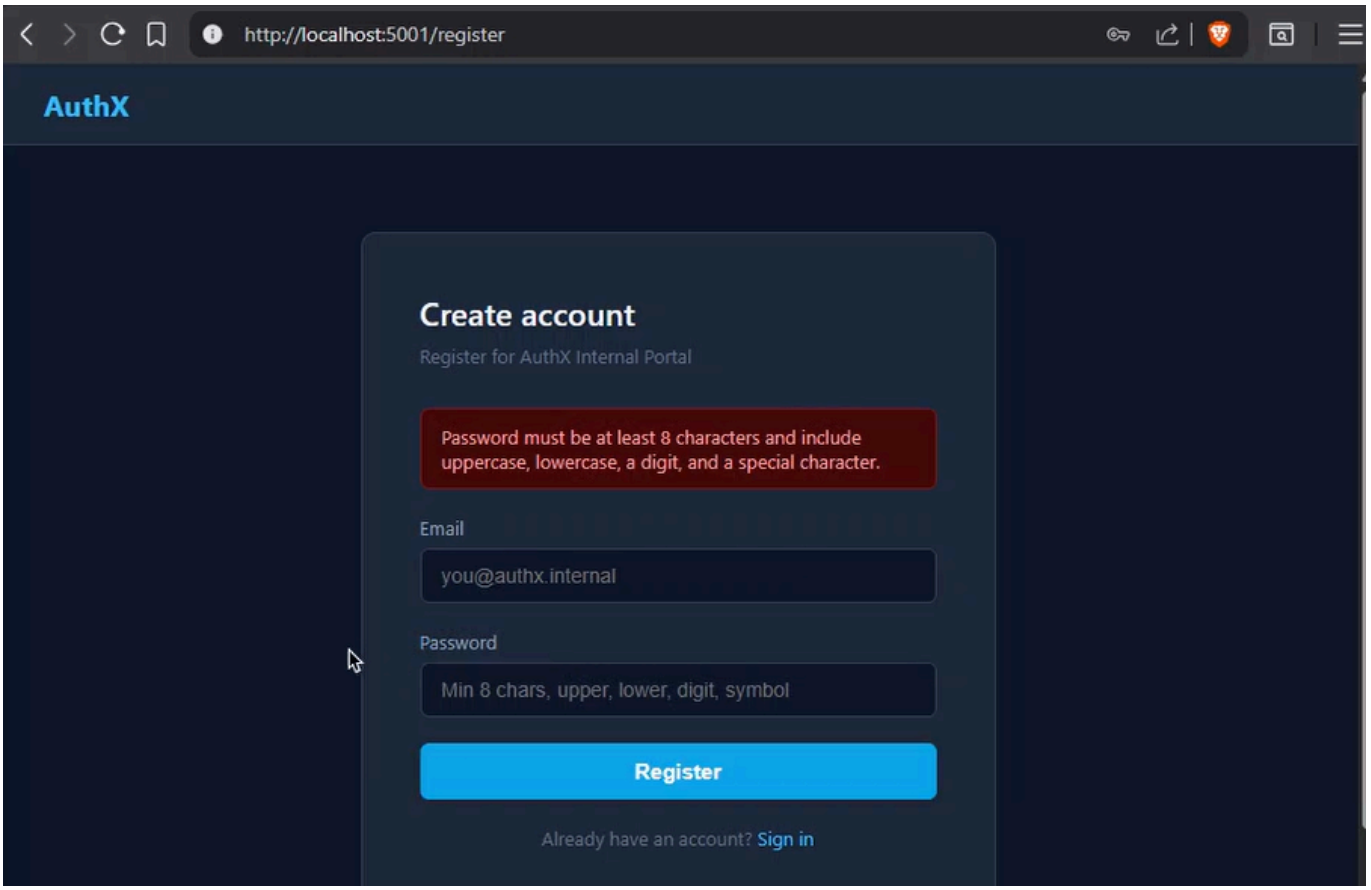
3.1 Register (`/register`)

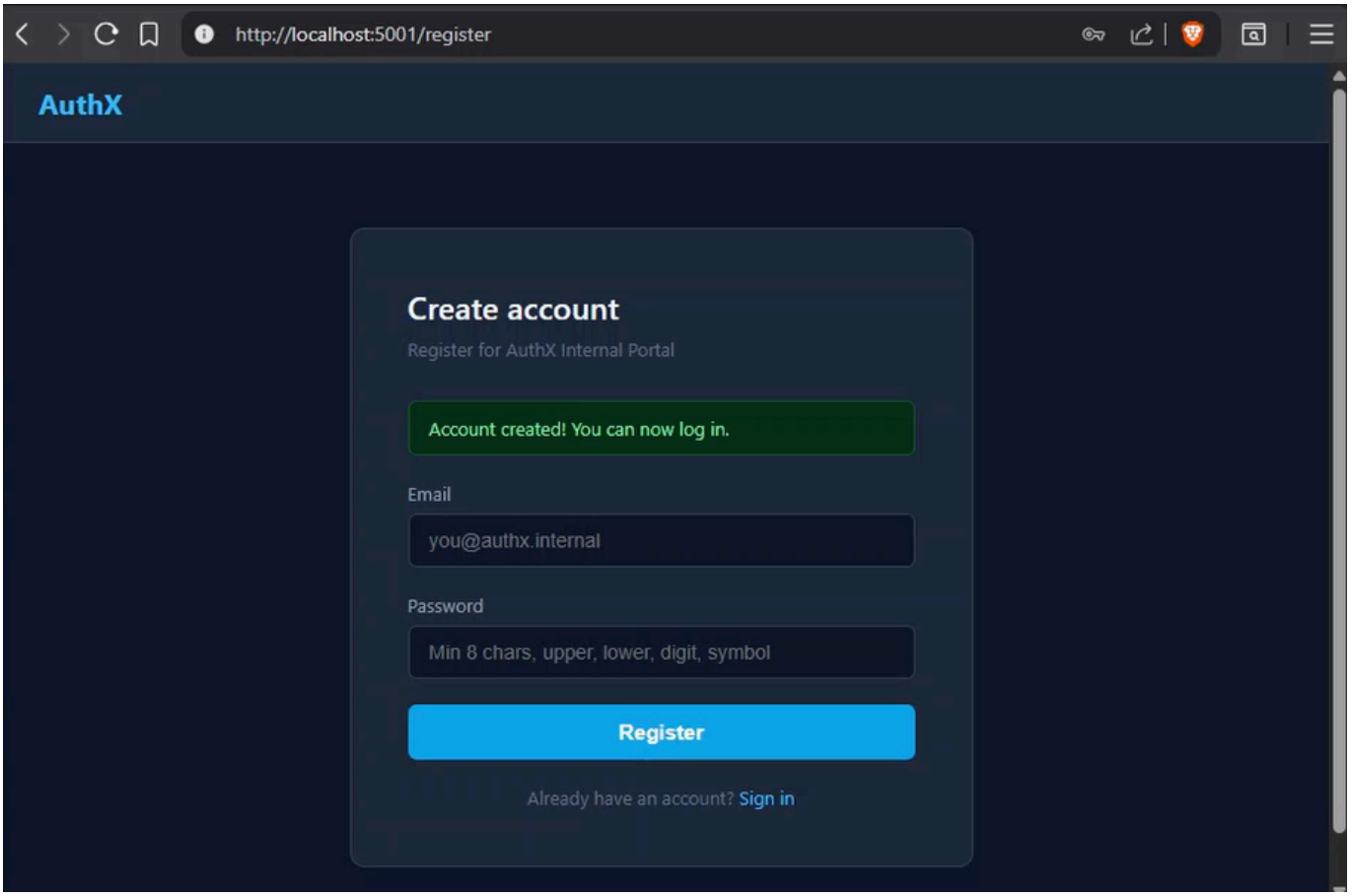
Accepts `email` and `password` via POST form. Validates that fields are present, inserts user into `users` table, records a `REGISTER` event in `audit_logs`.

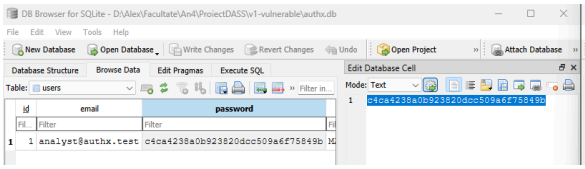
v1 flaw: No password complexity check. Password stored as MD5 hash.
v2 fix: Strong password policy enforced. bcrypt with auto-generated salt.

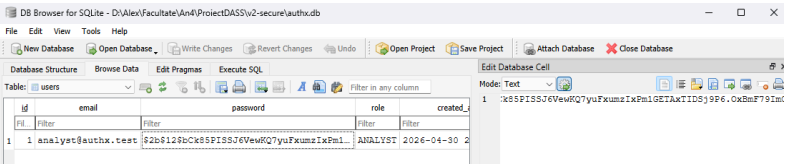


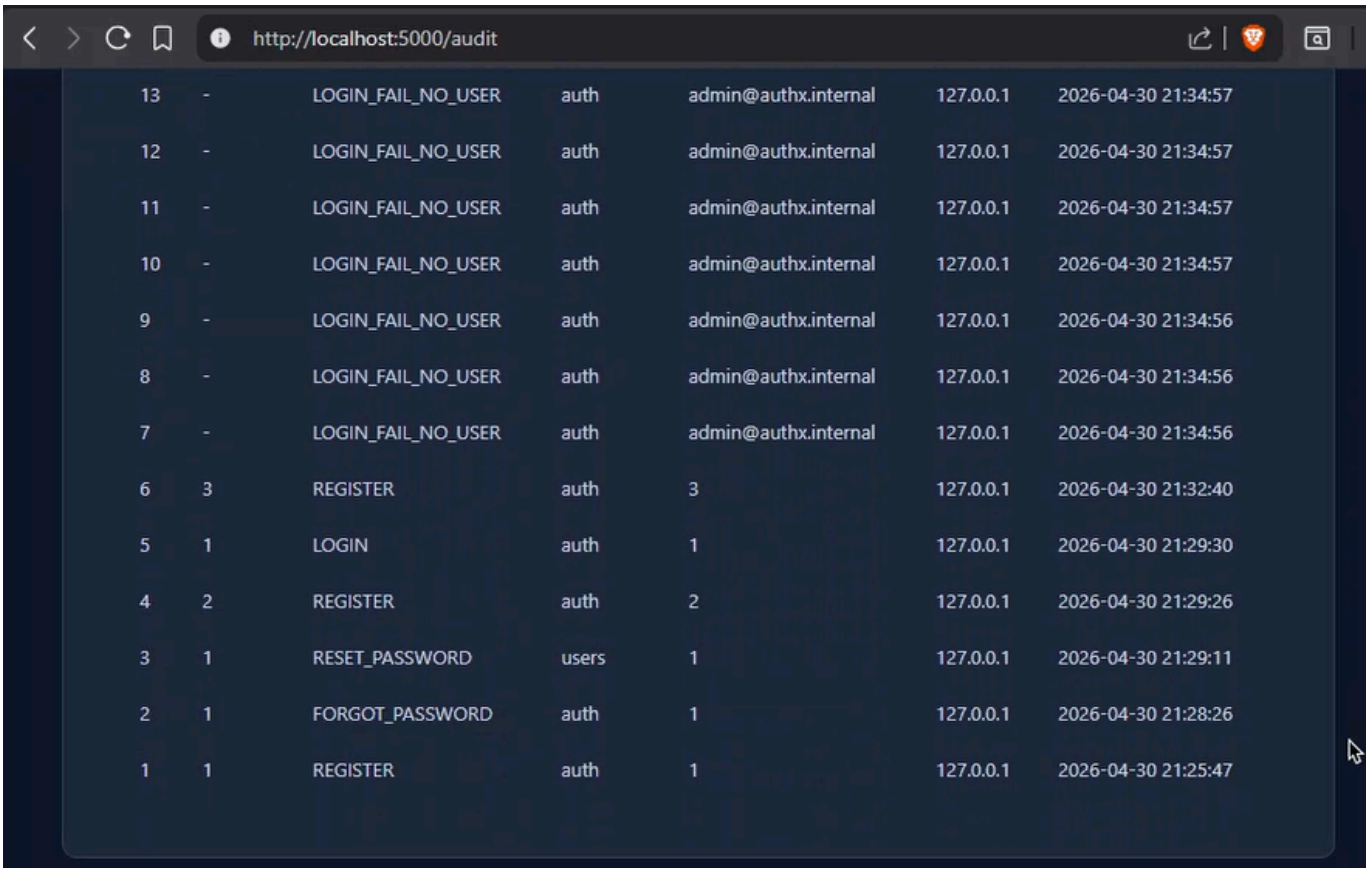



http://localhost:5001/register
AuthX
Create account
Register for AuthX Internal Portal
Password must be at least 8 characters and include uppercase, lowercase, a digit, and a special character.
Email
you@authx.internal
Password
Min 8 chars, upper, lower, digit, symbol
Register
Already have an account? Sign in


http://localhost:5001/register
AuthX
Create account
Register for AuthX Internal Portal
Account created! You can now log in.
Email
you@authx.internal
Password
Min 8 chars, upper, lower, digit, symbol
Register
Already have an account? Sign in


DB Browser for SQLite - D:\Alex\Facultate\An4\ProiectDASS\v1-vulnerable\authx.db
Table: users
id
email
password
analyst@authx.test
c4ca4238a0b923820dcc509a6f75849b


DB Browser for SQLite - D:\Alex\Facultate\An4\ProiectDASS\v2-secure\authx.db
Table: users
id
email
password
role
analyst@authx.test
ANALYST
2026-04-30

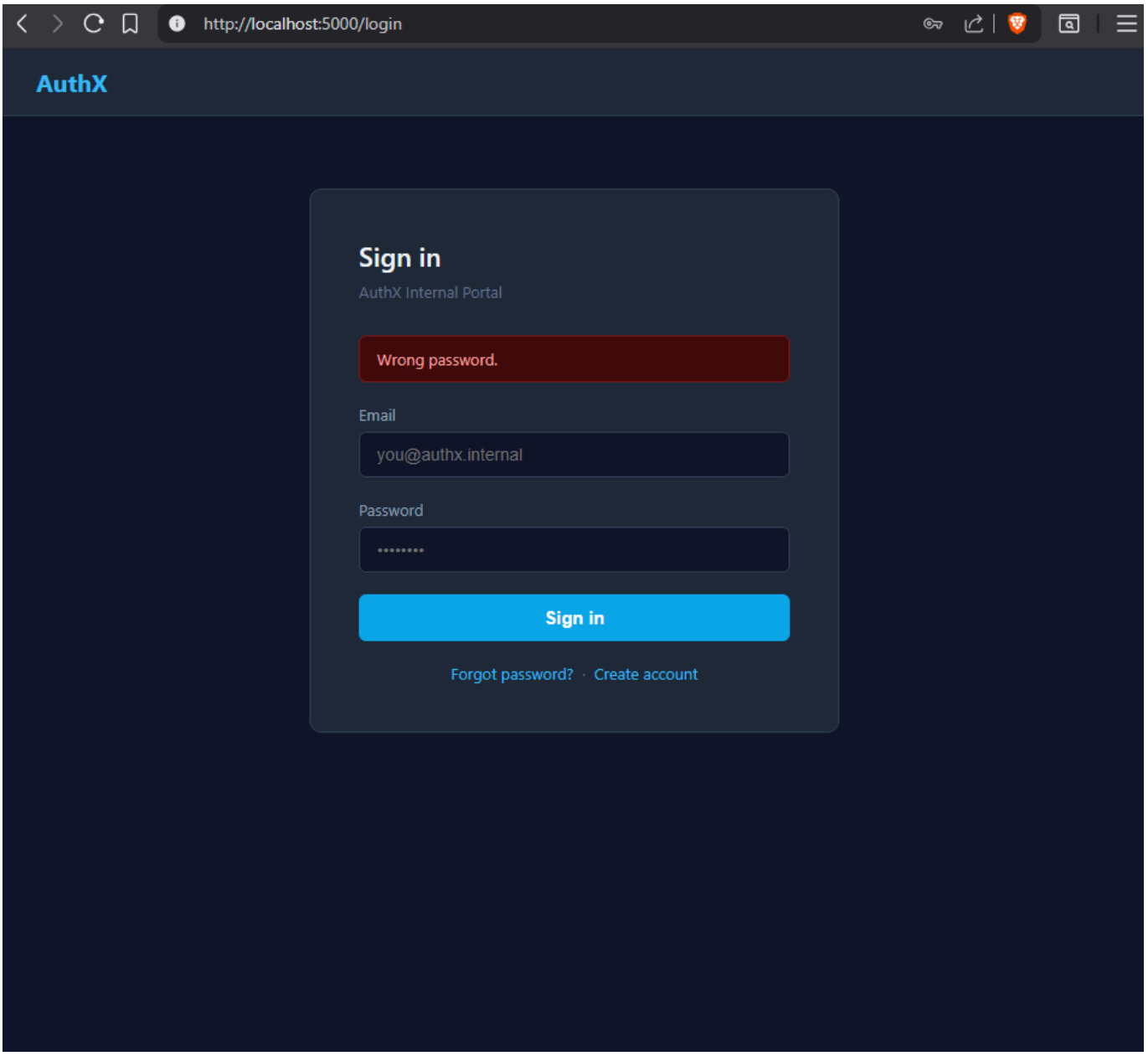


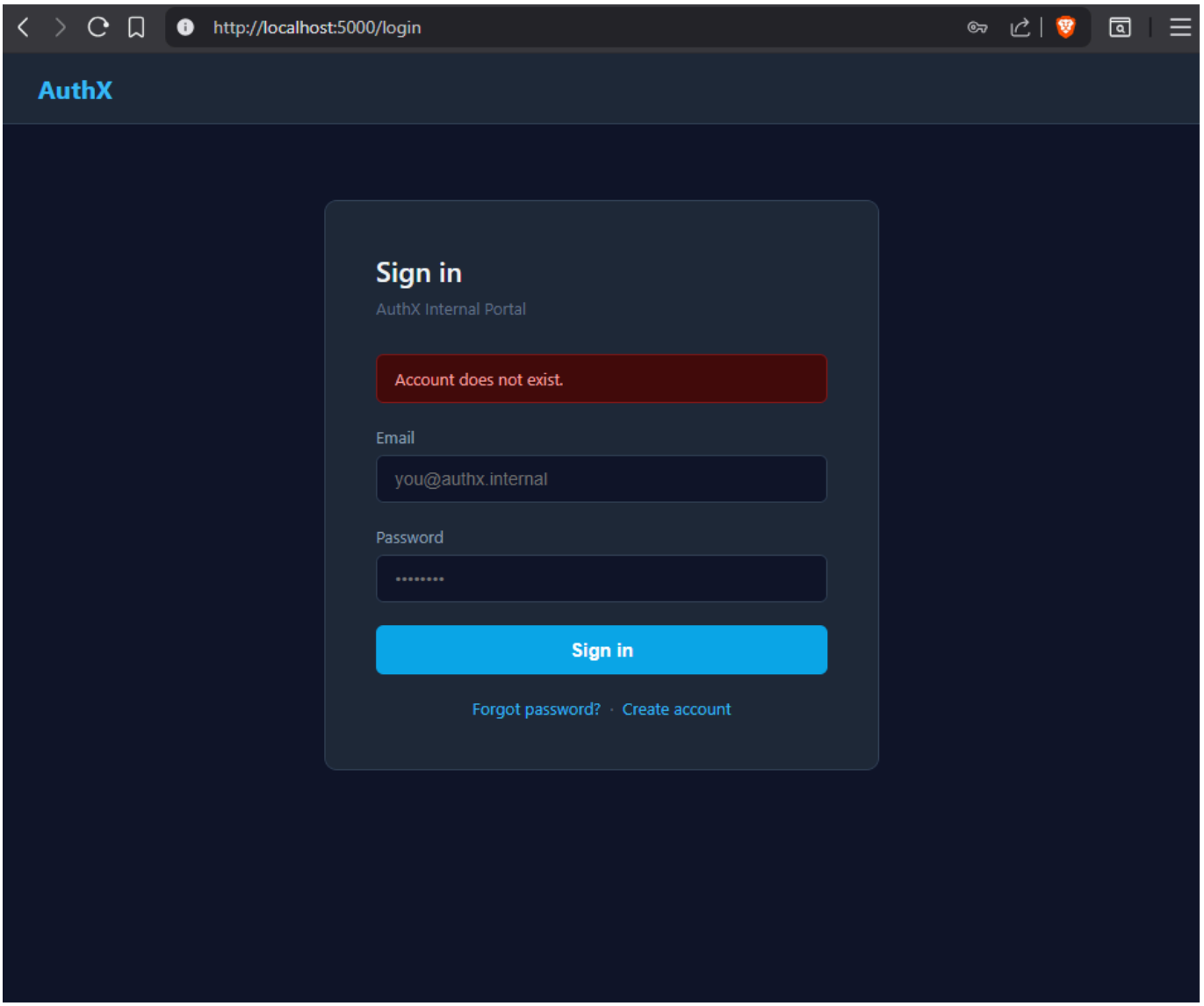
13	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:57
12	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:57
11	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:57
10	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:57
9	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:56
8	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:56
7	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:34:56
6	3	REGISTER	auth	3	127.0.0.1	2026-04-30 21:32:40
5	1	LOGIN	auth	1	127.0.0.1	2026-04-30 21:29:30
4	2	REGISTER	auth	2	127.0.0.1	2026-04-30 21:29:26
3	1	RESET_PASSWORD	users	1	127.0.0.1	2026-04-30 21:29:11
2	1	FORGOT_PASSWORD	auth	1	127.0.0.1	2026-04-30 21:28:26
1	1	REGISTER	auth	1	127.0.0.1	2026-04-30 21:25:47

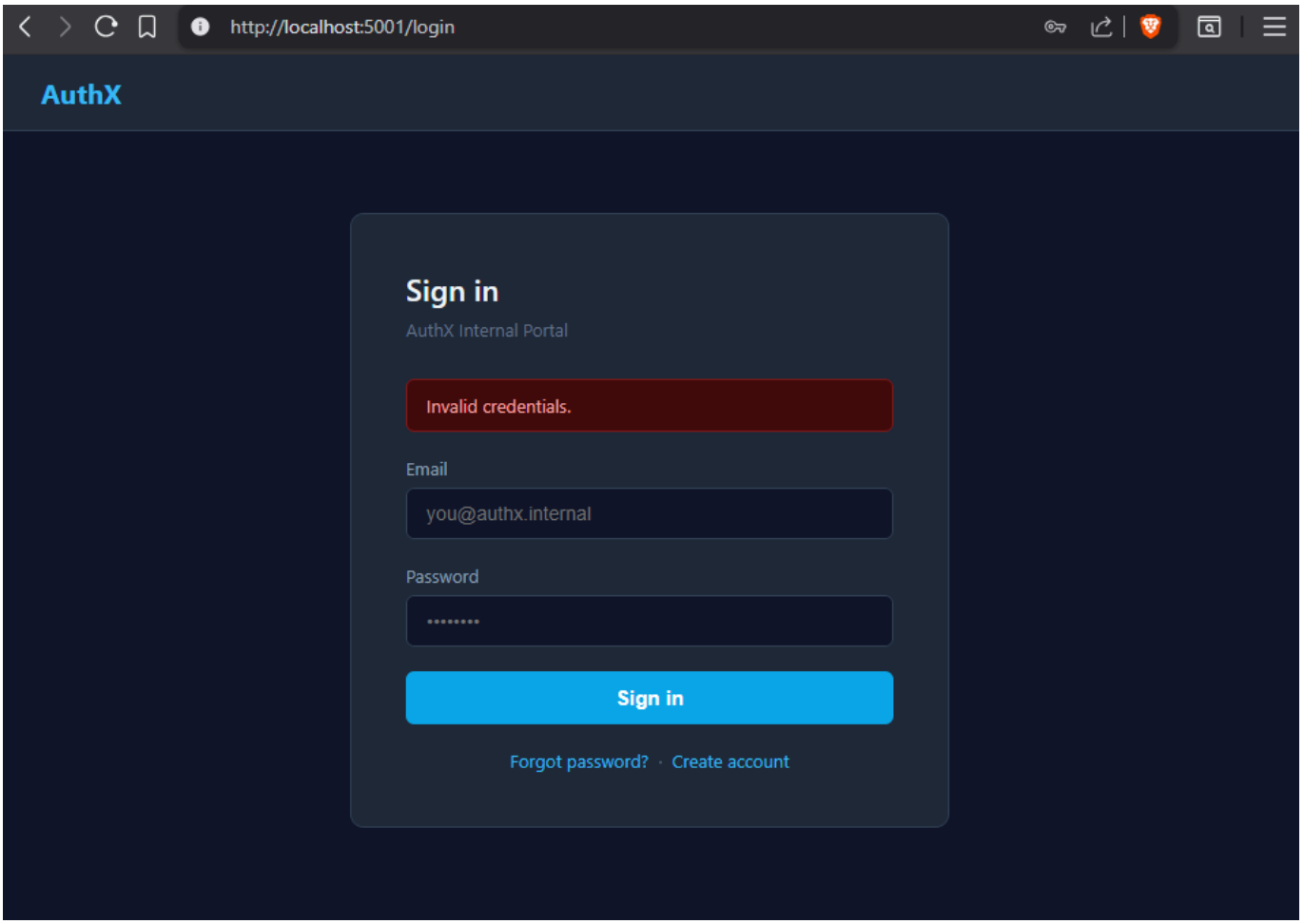
3.2 Login (`/login`)

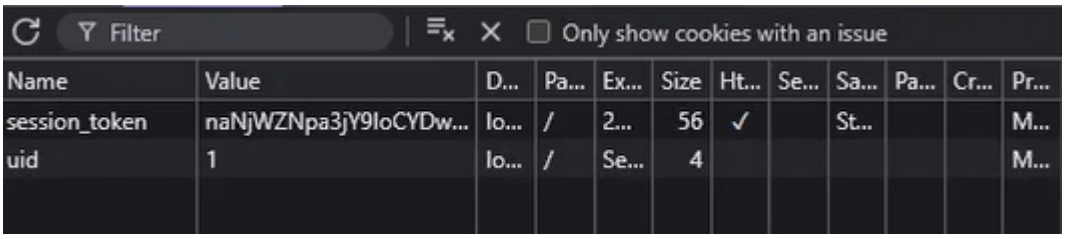
Accepts `email` + `password`. Looks up user in DB, compares password hash, sets a session cookie on success.

v1 flaw: MD5 comparison, different error messages for missing user vs. wrong password, cookie is just the user `id`.
v2 fix: bcrypt comparison, generic "Invalid credentials" for all failures, random session token with security flags.


http://localhost:5000/login
AuthX
Sign in
AuthX Internal Portal
Wrong password.
Email
you@authx.internal
Password
Sign in
Forgot password? · Create account

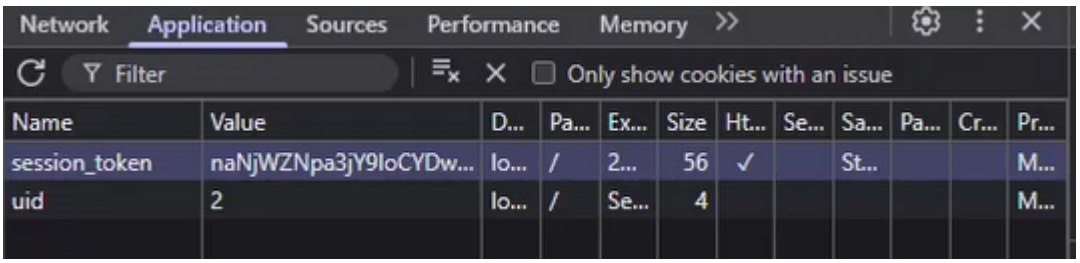

http://localhost:5000/login
AuthX
Sign in
AuthX Internal Portal
Account does not exist.
Email
you@authx.internal
Password
Sign in
Forgot password? · Create account

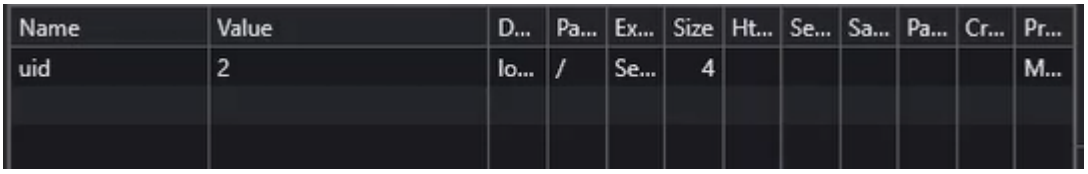




3.3 Logout (`/logout`)

Clears the session cookie. In v2, also removes the token from the server-side `sessions` dict so the token cannot be reused.

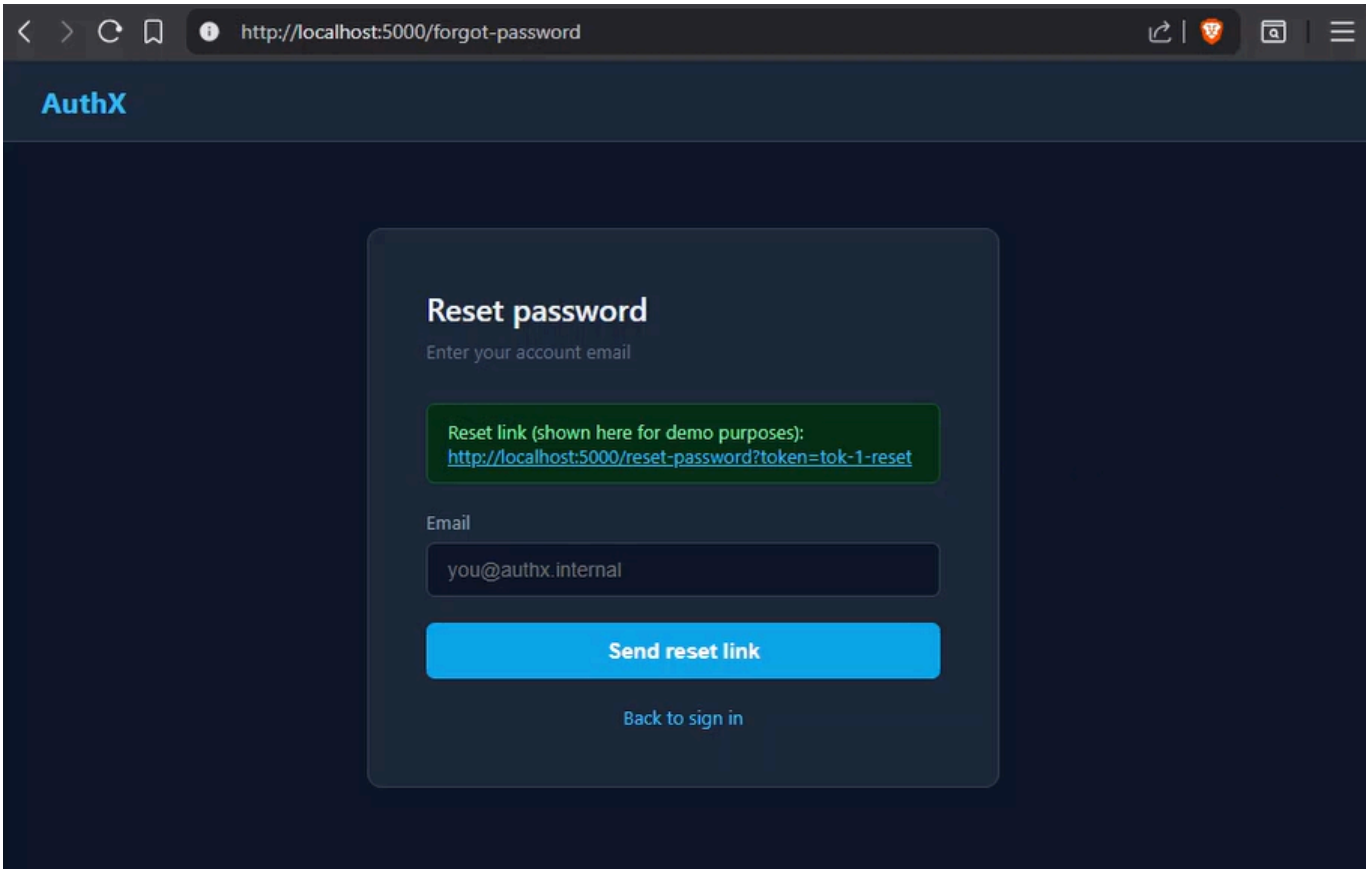


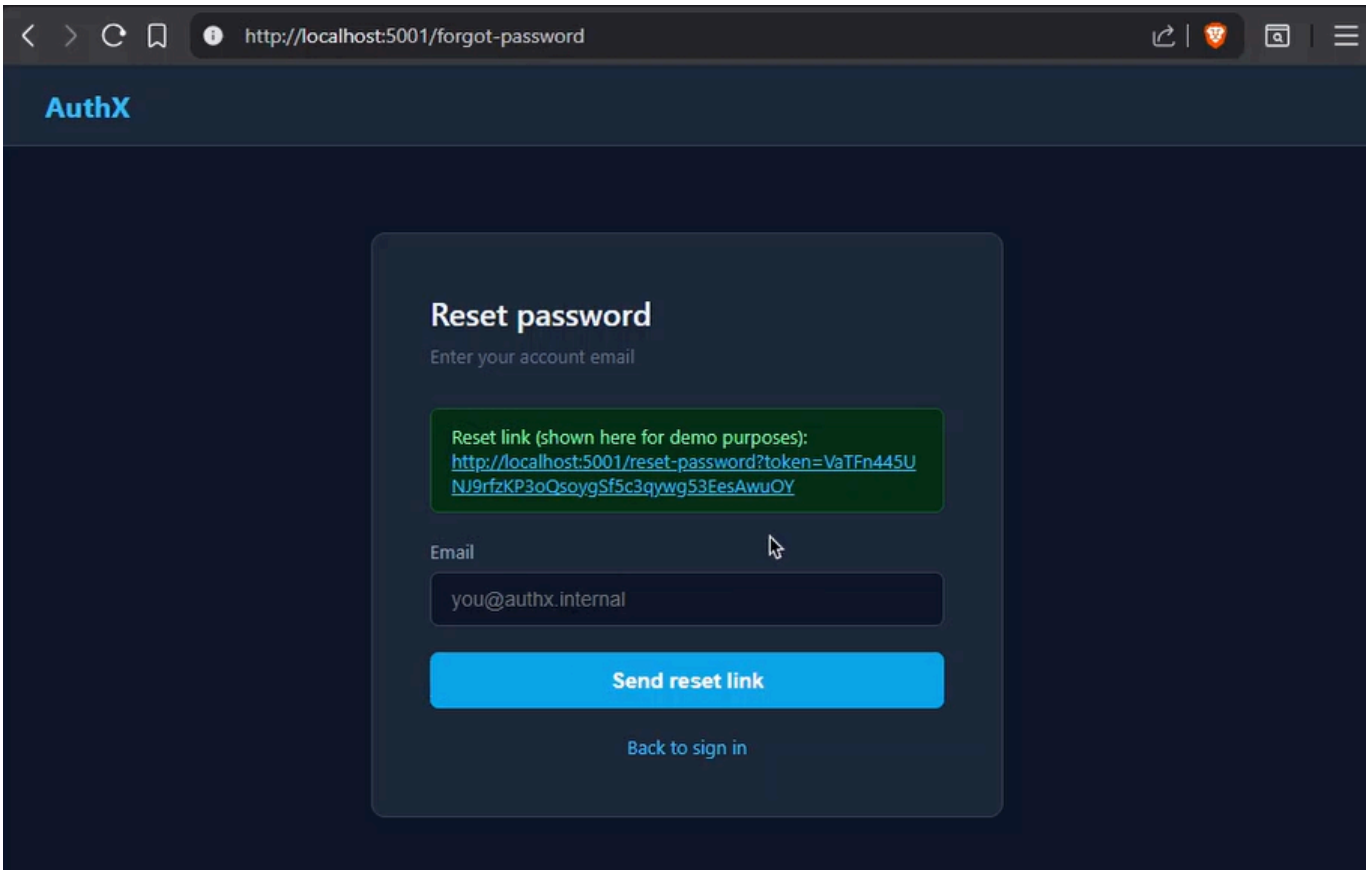


3.4 Password Reset (`/forgot-password`)

User submits email → app generates a reset token → link shown on page (simulating email delivery).

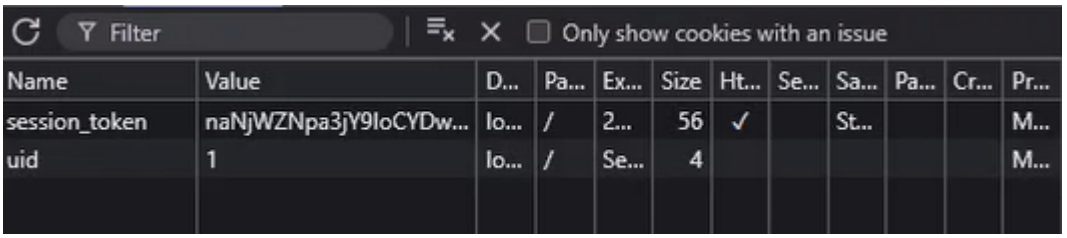
v1 flaw: Token is `tok-{user_id}-reset`, reusable, no expiry.
v2 fix: `secrets.token_urlsafe(32)`, 15-minute expiry, invalidated after first use.





3.5 Session Management

v1 uses a plain `uid` cookie (the integer user id).
v2 uses a 32-byte random token stored in a server-side dict, transmitted via `HttpOnly; SameSite=Strict; max-age=1800` cookie.



3.6 Ticket CRUD (`/tickets`, `/tickets/<id>/edit`, `/tickets/<id>/delete`)

The tickets page supports the full create-read-update-delete lifecycle:

Operation	Endpoint	Method
Create	`POST /tickets`	Form submission on the ticket list page
Read	`GET /tickets`	Lists all (MANAGER) or own (ANALYST) tickets; supports `?q=` search
Update	`POST /tickets/<id>/edit`	Edit form pre-filled with current values; title, description, severity, status
Delete	`POST /tickets/<id>/delete`	Removes the ticket and redirects back to the list

v1 flaw: No ownership check on update/delete — IDOR allows any authenticated user to modify or delete any ticket.
v2 fix: Ownership enforced before every write; only the ticket owner or a MANAGER may edit or delete.

---

4. Vulnerability Overview

#	Category	OWASP Reference	v1 Flaw
4.1	Weak Password Policy	A07:2021 - Identification and Authentication Failures	Any password accepted
4.2	Insecure Password Storage	A02:2021 - Cryptographic Failures	MD5 without salt
4.3	Brute Force / No Rate Limiting	A07:2021 - Identification and Authentication Failures	Unlimited login attempts
4.4	User Enumeration	A07:2021 - Identification and Authentication Failures	Distinct error messages

#	Category	OWASP Reference	v1 Flaw
4.5	Insecure Session Management	A07:2021 - Identification and Authentication Failures	Predictable cookie, no flags
4.6	Insecure Password Reset	A07:2021 - Identification and Authentication Failures	Predictable, reusable token
4.7	IDOR on Tickets	A01:2021 - Broken Access Control	No ownership check on edit/delete

---

5. Proof of Concept - Attacks

5.1 Weak Password Policy

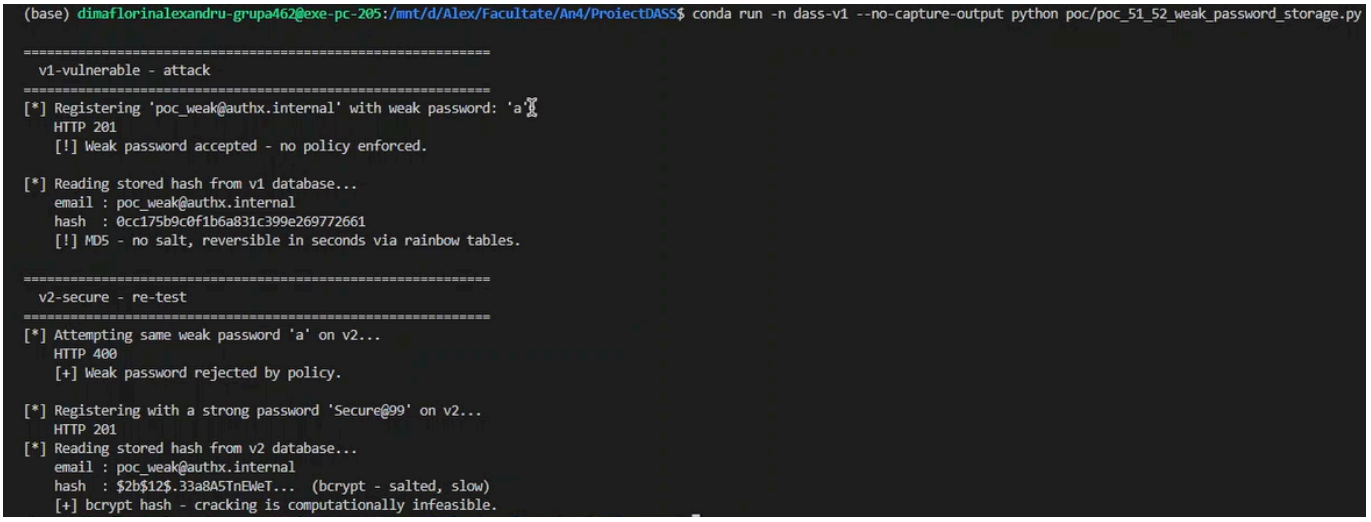
Vulnerability: v1 accepts passwords of any length and complexity, including single characters.

PoC script: `poc/poc_51_52_weak_password_storage.py`

```
cd v1-vulnerable
python app.py            # terminal 1
# terminal 2:
python ../poc/poc_51_52_weak_password_storage.py
```

Expected output:

```
[*] Registering with weak password: 'a'
    HTTP 201
[*] Reading password hash directly from database...
    email         : poc_weak@authx.internal
    password hash: 60b725f10c9c85c70d97880dfe8191b3
[!] MD5 hash is fast and unsalted - trivially reversible with any online cracker.
```



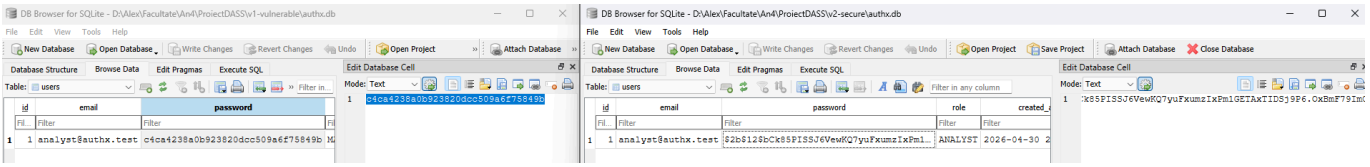
Impact: An attacker who obtains the database can crack all passwords instantly with rainbow tables or online MD5 lookup services.

5.2 Insecure Password Storage

Vulnerability: Passwords are hashed with MD5, which is a fast, unsalted algorithm. MD5 has been broken for password storage purposes for over a decade.

```
# v1 - insecure
def weak_hash(password: str) -> str:
    return hashlib.md5(password.encode()).hexdigest()
```

PoC: Same script as 4.1 - the database dump shows the raw MD5 hash.



5.3 Brute Force / No Rate Limiting

Vulnerability: v1 has no attempt counter, lockout mechanism or delay. An attacker can make thousands of requests per second.

PoC script: `poc/poc_53_brute_force.py`

```
python ../poc/poc_53_brute_force.py
```

Expected output (abbreviated):

```
[*] Brute-forcing admin@authx.internal with 15 passwords...

    123456              -> fail (401)
    password            -> fail (401)
    qwerty              -> fail (401)
    ...
[!] All requests completed without any block or lockout.
```

```
(base) dimaflorinalexandru-grupa462@exe-pc-205:/mnt/d/Alex/Facultate/An4/ProiectDASS$ conda run -n dass-v1 --no-capture-output python poc/poc_53_brute_force.py

============================================================
  v1-vulnerable - attack
============================================================
[*] Sending 15 login attempts to v1...

    123456              -> fail (401)
    password            -> fail (401)
    qwerty              -> fail (401)
    letmein             -> fail (401)
    admin               -> fail (401)
    welcome             -> fail (401)
    monkey              -> fail (401)
    dragon              -> fail (401)
    master              -> fail (401)
    abc123              -> fail (401)
    pass1234            -> fail (401)
    iloveyou            -> fail (401)
    sunshine            -> fail (401)
    princess            -> fail (401)
    football            -> fail (401)

    [!] All attempts completed - no lockout triggered.

============================================================
  v2-secure - re-test
============================================================
[*] Sending 15 login attempts to v2...

    123456              -> fail (401)
    password            -> fail (401)
```

5.4 User Enumeration

Vulnerability: v1 returns `"Account does not exist."` for an unknown email and `"Wrong password."` for a wrong password on an existing account. This lets an attacker distinguish valid emails from invalid ones.

PoC script: `poc/poc_54_user_enumeration.py`

```
python ../poc/poc_54_user_enumeration.py
```

Expected output:

```
[*] Probing with an email that does NOT exist:
    HTTP 401  |  "Account does not exist."

[*] Probing with an email that EXISTS (wrong password):
    HTTP 401  |  "Wrong password."

[!] Different messages allow enumeration of valid accounts.
```

Impact: Attacker builds a list of valid usernames, then targets only those accounts in a credential-stuffing or brute-force attack.

5.5 Insecure Session Management

Vulnerability: After login, v1 sets a cookie `uid=<integer>`, which:

- Is the user's database ID (predictable)
- Has no `HttpOnly` flag (accessible from JavaScript - XSS risk)

- Has no `Secure` flag (sent over plain HTTP)
- Has no `SameSite` attribute (CSRF risk)
- Never expires

PoC script: `poc/poc_55_insecure_session.py`

```
python ../poc/poc_55_insecure_session.py
```

Expected output:

```
[*] Logging in with legitimate credentials...
    Set-Cookie uid = '1'
    Cookie has no HttpOnly, Secure or SameSite flags.

[*] Forging session for user id=1 (likely the first registered admin)...
    HTTP 200
    [!] Dashboard loaded for user id=1 without knowing the password!
```

```
(base) dimaflorinalexandru-grupa462@exe-pc-205:/mnt/d/Alex/Facultate/An4/ProiectDASS$ conda run -n dass-v1 --no-capture-output python poc/poc_55_insecure_session.py

============================================================
  v1-vulnerable - attack
============================================================
[*] Logging in as 'analyst@authx.internal' to observe the cookie...
    Cookie received: uid=None
    Value is the plain integer user id - predictable and forgeable.

[*] Forging a session for user id=1 without knowing their password...
    HTTP 200
    [!] Dashboard returned for user id=1 with a forged cookie.

============================================================
  v2-secure - re-test
============================================================
[*] Forging session_token='fake-token-123' on v2...
    HTTP 302
    [+] Fake token rejected - redirected to login.
```

Network | Application | Sources | Performance | Memory

Filter | Only show cookies with an issue

Name	Value	D...	Pa...	Ex...	Size	Ht...	Se...	Sa...	Pa...	Cr...	Pr...
session_token	naNjWZNpa3jY9IoCYDw...	lo...	/	2...	56	✓		St...			M...
uid	2	lo...	/	Se...	4						M...

5.6 Insecure Password Reset

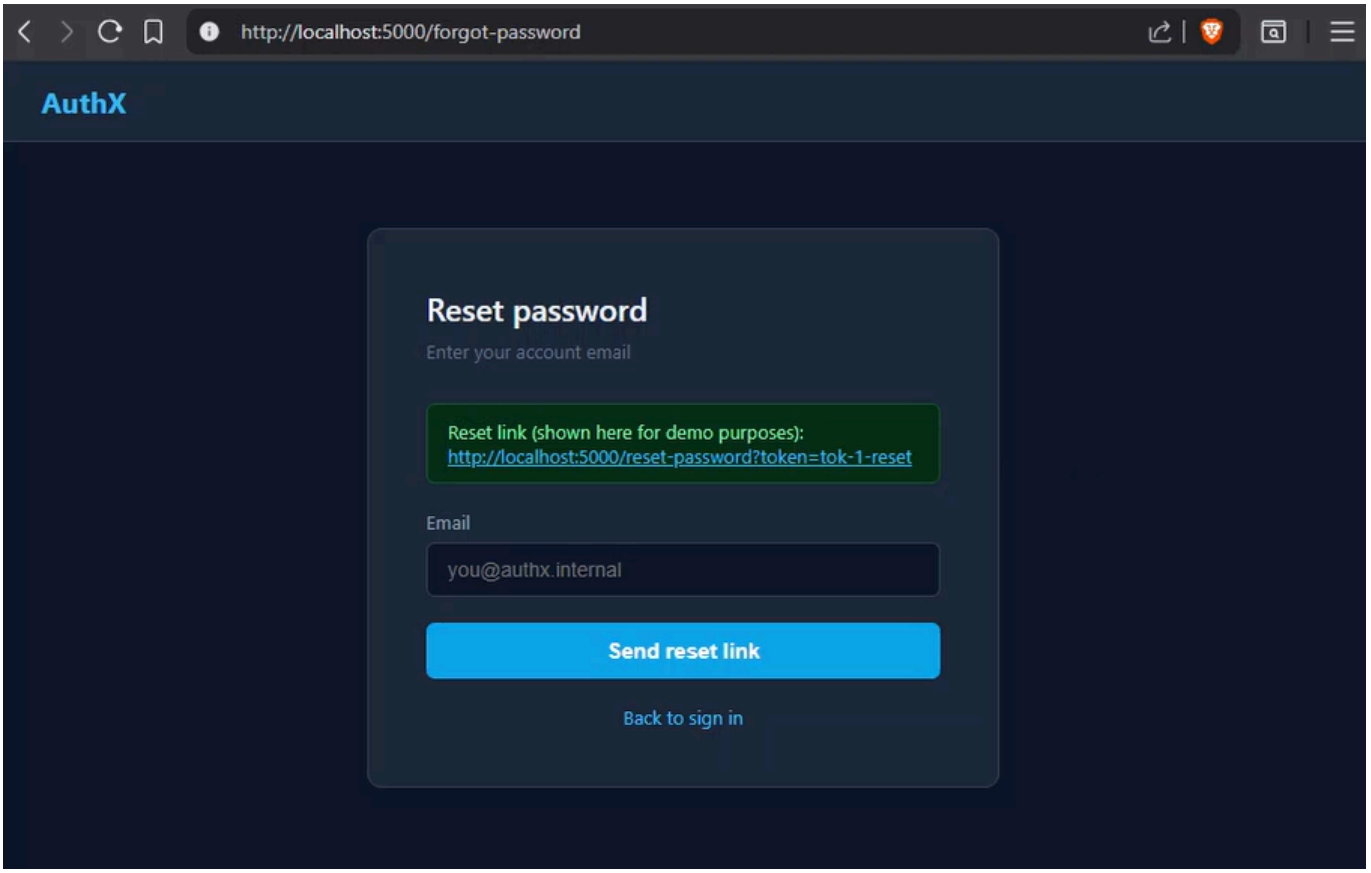
Vulnerability: The reset token is `tok-{user_id}-reset` - directly derived from the user id. It has no expiry and can be used multiple times.

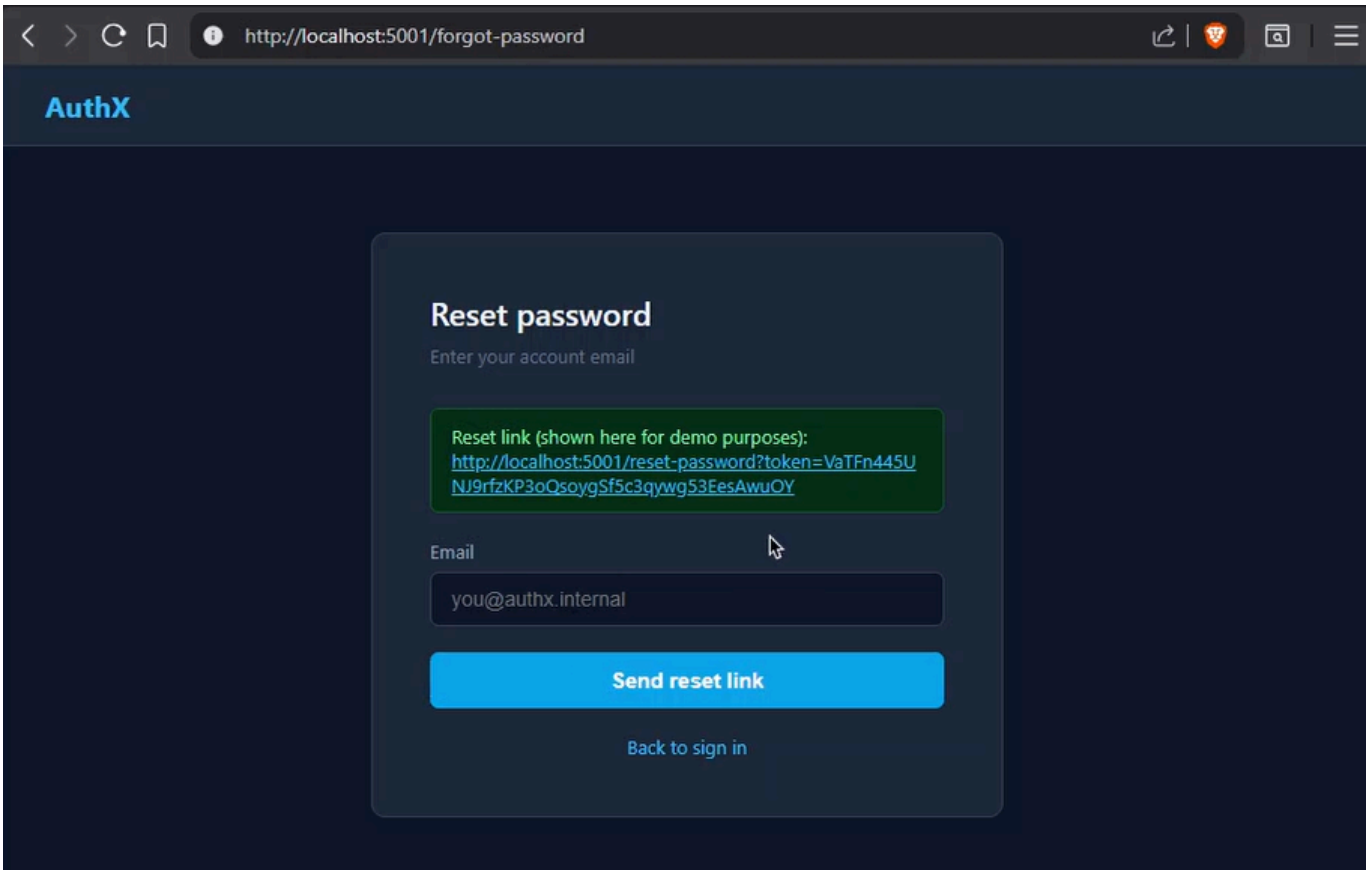
PoC script: `poc/poc_56_reset_token.py`

```
python ../poc/poc_56_reset_token.py
```

Expected output:

```
[*] Guessed token for user id=1: 'tok-1-reset'
[*] First reset attempt...
    HTTP 200
    [!] Password successfully changed without owning the account!

[*] Second reset attempt with the same token (reuse test)...
    HTTP 200
    [!] Token accepted a second time - it is reusable (no invalidation).
```





5.7 IDOR on Ticket CRUD

Vulnerability: v1's edit and delete endpoints accept a ticket `id` from the URL path without verifying that the requesting user owns that ticket. Any authenticated user can modify or delete any ticket by crafting a request to `/tickets/<id>/edit` or `/tickets/<id>/delete`.

PoC: Register two accounts (User A and User B). Log in as User A and create a ticket. Note its id. Log out and log in as User B. Send:

```
POST /tickets/<user_a_ticket_id>/delete
```

Impact: Any authenticated employee can tamper with or destroy another user's tickets, breaking data integrity and confidentiality.

---

6. Impact Analysis

Vulnerability	What an attacker gains
4.1 Weak policy	Accounts protected by trivial passwords; exploitable immediately
4.2 MD5 storage	Full plaintext passwords recovered from a database dump in seconds
4.3 No rate limiting	Automated password guessing against any account
4.4 User enumeration	Valid email list for targeted attacks, phishing, credential stuffing
4.5 Insecure session	Account takeover by guessing/forging cookie; XSS can steal it

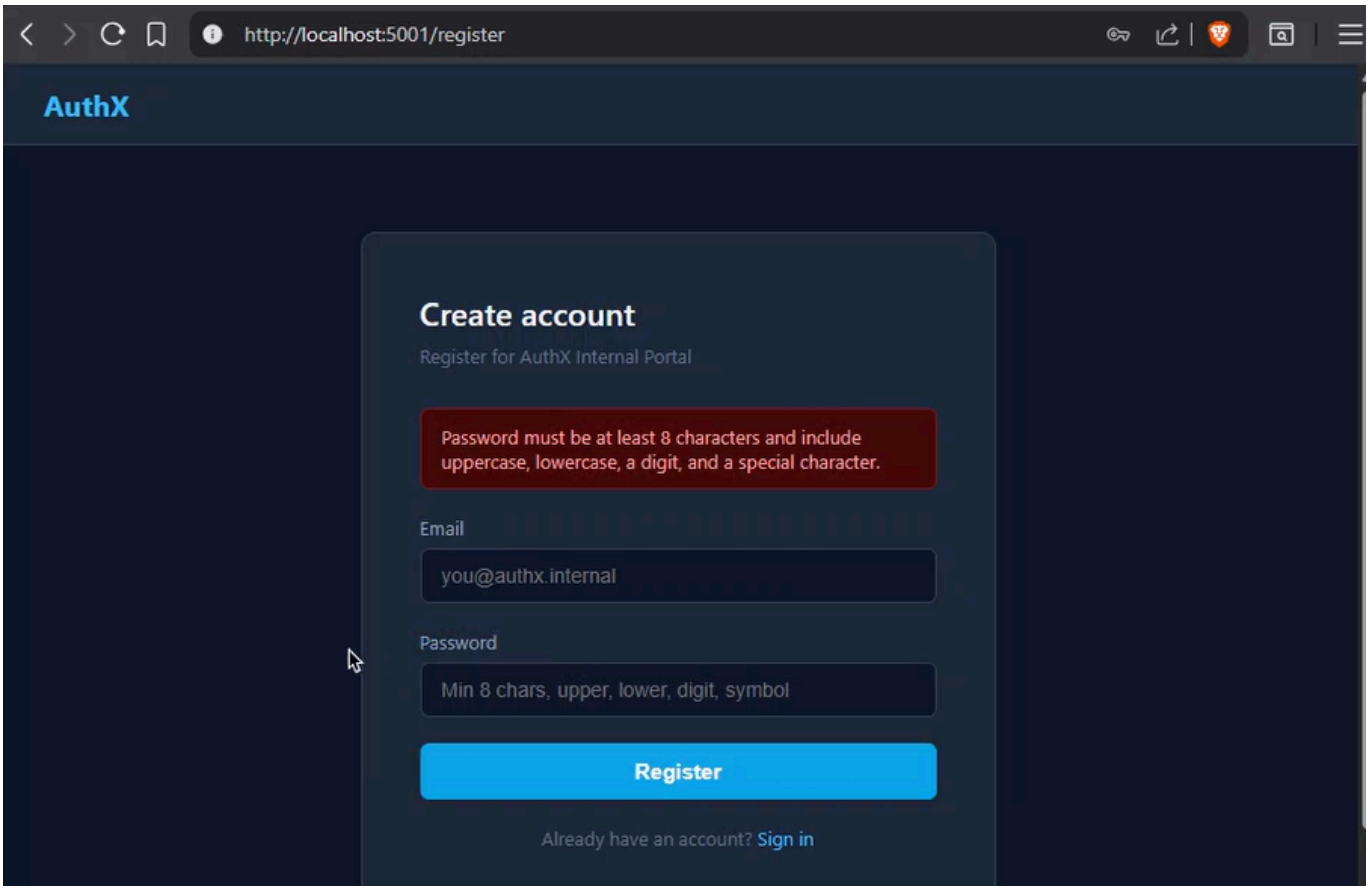
Vulnerability	What an attacker gains
4.6 Weak reset token	Account takeover for any user by guessing their reset token
4.7 IDOR on tickets	Any authenticated user can edit or delete another user's tickets

---

7. Fixes Implemented (v2)

7.1 - Strong Password Policy

```
def is_strong_password(password: str) -> bool:
    return (
        len(password) >= 8
        and bool(re.search(r"[A-Z]", password))
        and bool(re.search(r"[a-z]", password))
        and bool(re.search(r"\d", password))
        and bool(re.search(r'[!@#$%^&*()\-_=+\[\]{}|;:,.<>?/]', password))
    )
```



7.2 - bcrypt with Salt

```
hashed = bcrypt.hashpw(password.encode(), bcrypt.gensalt()).decode()
```

7.3 - Rate Limiting & Account Lockout

After 5 failed attempts the account is blocked for 5 minutes. All failed attempts are written to `audit_logs`.

7.4 - Generic Error Message + Timing Equalisation

```
# Always run bcrypt to prevent timing oracle
dummy = bcrypt.hashpw(b"__dummy__", bcrypt.gensalt())
if user:
    valid = bcrypt.checkpw(password.encode(), user["password"].encode())
else:
    bcrypt.checkpw(password.encode(), dummy)
    valid = False

# Enforce minimum 0.5 s response time
elapsed = time.monotonic() - start
if elapsed < 0.5:
    time.sleep(0.5 - elapsed)
```

Single generic error: `"Invalid credentials."`

7.5 - Secure Session Cookie

```
resp.set_cookie(
    "session_token",
    session_token,
    httponly=True,
    secure=False,       # set True when serving over HTTPS
    samesite="Strict",
    max_age=1800,
)
```

- Token: `secrets.token_urlsafe(32)` (256-bit random)
- Server-side `sessions` dict; cleared on logout
- 30-minute expiry

```
(base) dimaflorinalexandru-grupa462@exe-pc-205:/mnt/d/Alex/Facultate/An4/ProiectDASS$ conda run -n dass-v1 --no-capture-output python poc/poc_55_insecure_session.py

============================================================
  v1-vulnerable - attack
============================================================
[*] Logging in as 'analyst@authx.internal' to observe the cookie...
    Cookie received: uid=None
    Value is the plain integer user id - predictable and forgeable.

[*] Forging a session for user id=1 without knowing their password...
    HTTP 200
    [!] Dashboard returned for user id=1 with a forged cookie.

============================================================
  v2-secure - re-test
============================================================
[*] Forging session_token='fake-token-123' on v2...
    HTTP 302
    [+] Fake token rejected - redirected to login.
```

7.6 - Secure One-Time Reset Token

```
token = secrets.token_urlsafe(32)
reset_tokens[token] = {
    "user_id": user["id"],
    "expires": datetime.now() + timedelta(minutes=15),
}
```

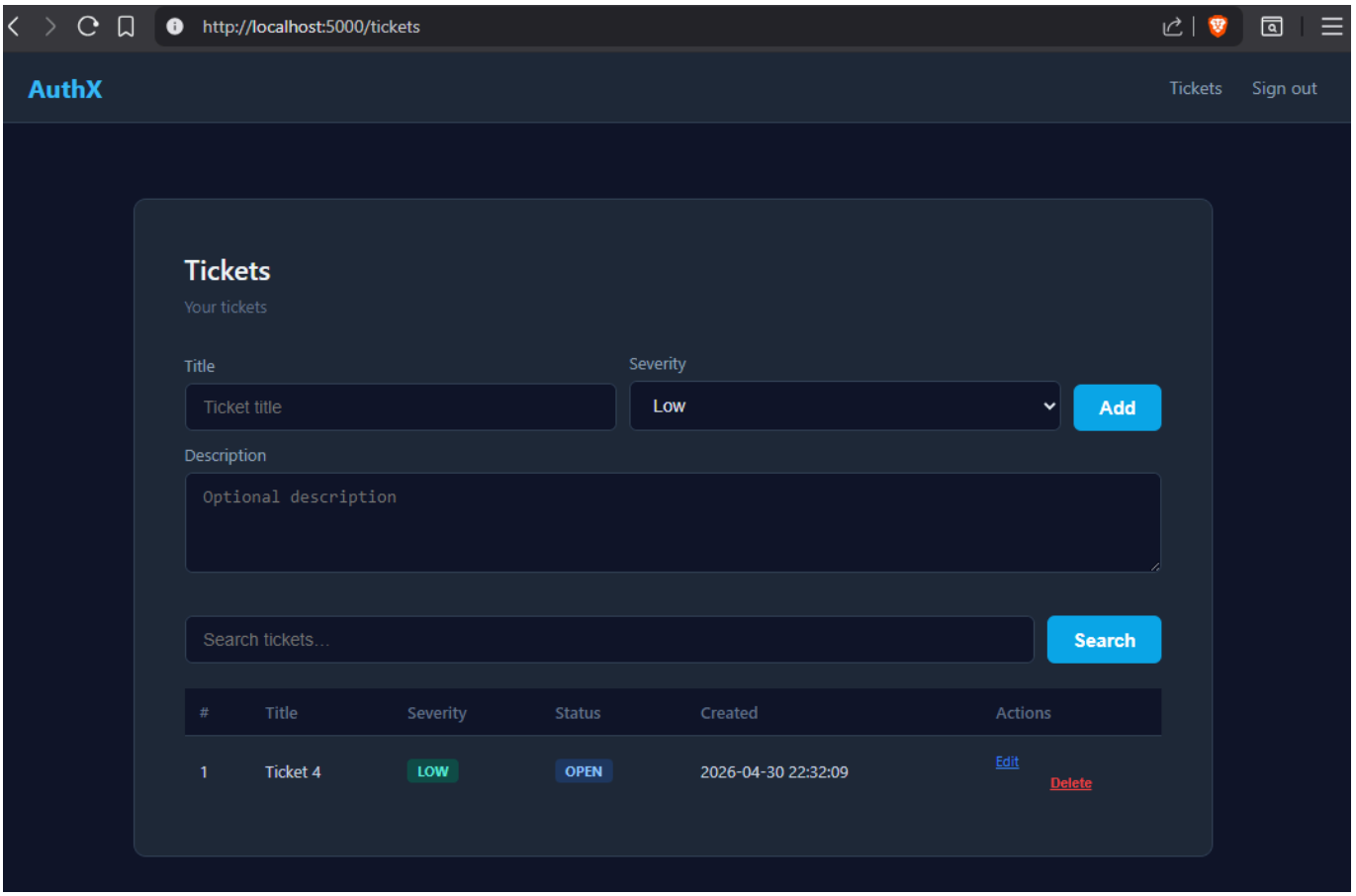
Token is deleted from `reset_tokens` on first use.

```
(base) dimaflorinalexandru-grupa462@exe-pc-205:/mnt/d/Alex/Facultate/An4/ProiectDASS$ conda run -n dass-v1 --no-capture-output python poc/poc_56_reset_token.py

============================================================
  v1-vulnerable - attack
============================================================
[*] Constructed token for user id=1: 'tok-1-reset'
[*] First reset attempt...
    HTTP 200
    [!] Password changed without owning the account.

[*] Second reset with the same token (reuse test)...
    HTTP 200
    [!] Token reused successfully - no invalidation on v1.

============================================================
  v2-secure - re-test
============================================================
[*] Trying the same predictable token format against v2...
    HTTP 400
    [+] Predictable token rejected by v2.
```

7.7 - Ownership Enforcement on Ticket CRUD

```
# Access control: only the owner or a MANAGER may edit/delete
if ticket["owner_id"] != user["id"] and user["role"] != "MANAGER":
    record_event(user["id"], "UNAUTHORIZED_TICKET_EDIT", "tickets",
str(ticket_id))
    return redirect(url_for("tickets"))
```

The check runs before any database write. Unauthorized attempts are silently redirected and recorded in `audit_logs`. The template additionally hides Edit/Delete buttons for rows the current user does not own (defence in depth).



8. Re-test After Fix

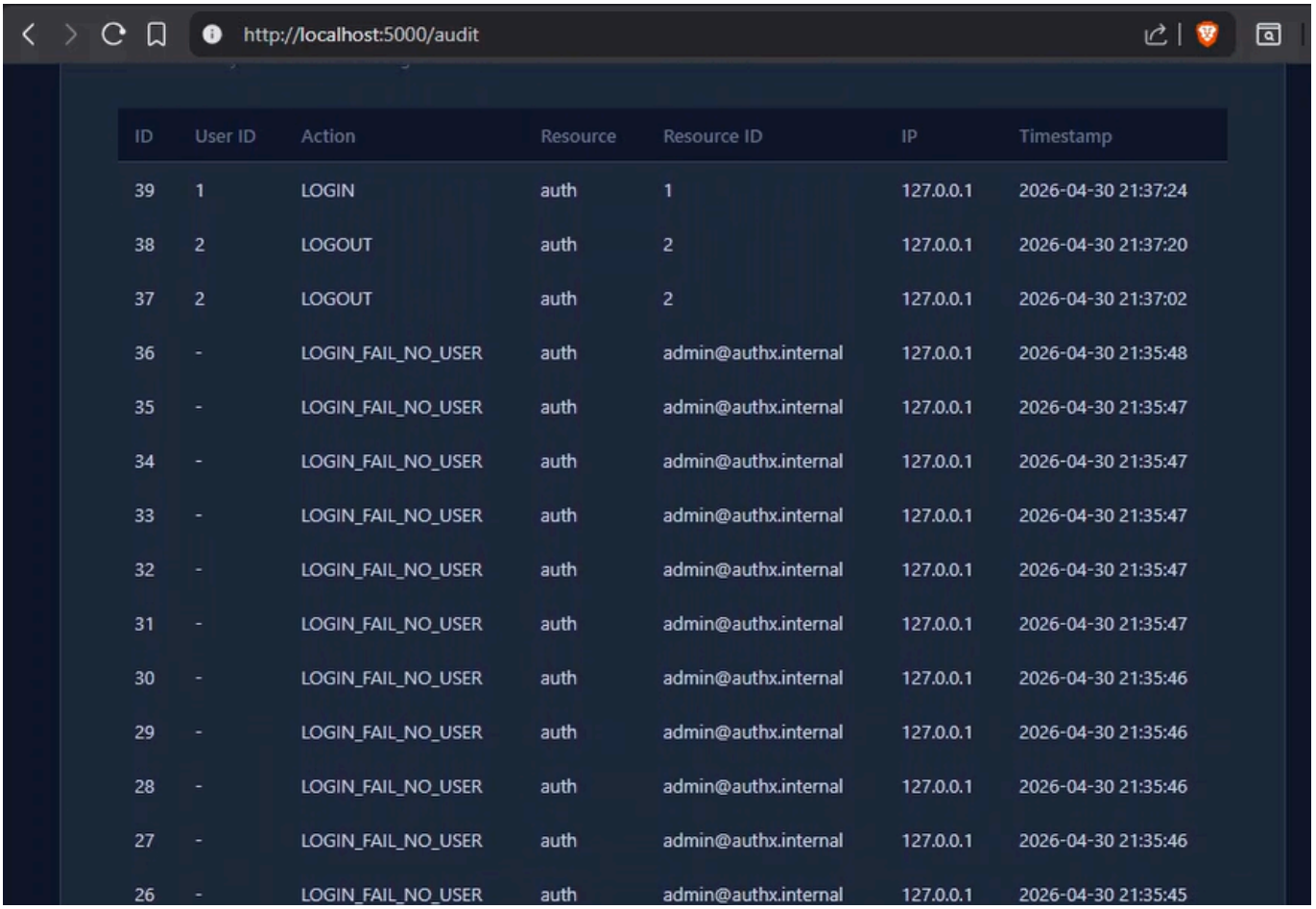
#	PoC script	v1 result	v2 result
5.1	`poc_51_52_weak_password_storage.py`	HTTP 201, password `a` accepted	HTTP 400, policy error
5.2	DB dump	MD5 hash visible	`$2b$...` bcrypt hash
5.3	`poc_53_brute_force.py`	All 15 requests succeed	Blocked after attempt 5 (HTTP 429)
5.4	`poc_54_user_enumeration.py`	Two distinct messages	Single generic message for both
5.5	`poc_55_insecure_session.py`	`uid=1` forges session	Random token; forged value rejected
5.6	`poc_56_reset_token.py`	Token reused successfully	Second use returns 400
5.7	Manual (browser / curl)	Delete User A's ticket as User B	Silently redirected; audit event logged

9. Audit & Logging

Every significant action is recorded in `audit_logs` with the user id, action name, resource, and client IP address.

Logged actions: `LOGIN`, `LOGIN_FAIL`, `LOGIN_FAIL_NO_USER`, `LOGIN_FAIL_WRONG_PASS`, `REGISTER`, `LOGOUT`, `FORGOT_PASSWORD`, `RESET_PASSWORD`, `CREATE_TICKET`, `UPDATE_TICKET`, `DELETE_TICKET`, `UNAUTHORIZED_TICKET_EDIT` (v2 only), `UNAUTHORIZED_TICKET_DELETE` (v2 only).

Managers can view the log at `/audit`.



ID	User ID	Action	Resource	Resource ID	IP	Timestamp
39	1	LOGIN	auth	1	127.0.0.1	2026-04-30 21:37:24
38	2	LOGOUT	auth	2	127.0.0.1	2026-04-30 21:37:20
37	2	LOGOUT	auth	2	127.0.0.1	2026-04-30 21:37:02
36	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:48
35	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:47
34	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:47
33	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:47
32	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:47
31	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:47
30	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:46
29	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:46
28	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:46
27	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:46
26	-	LOGIN_FAIL_NO_USER	auth	admin@authx.internal	127.0.0.1	2026-04-30 21:35:45

---

10. Conclusions

The project demonstrated the complete vulnerability-attack-fix cycle for six common authentication weaknesses mapped to OWASP A07:2021 - Identification and Authentication Failures.

Key lessons:

- **Never store passwords in plaintext or with fast hashes** (MD5/SHA-1). bcrypt, scrypt or Argon2 must be used.
- **Rate limiting is a critical control** even for low-traffic applications; a simple counter with a lockout is sufficient.
- **Generic error messages prevent user enumeration** and should be the default for all authentication endpoints.

- **Session tokens must be random, server-validated, and tied to secure cookie attributes** - an integer user id is never an acceptable session identifier.
- **Password reset tokens are credentials** and must be treated as such: cryptographically random, short-lived, and single-use.
- **Access control must be enforced server-side on every write operation** — hiding UI elements is not a substitute. IDOR vulnerabilities arise whenever the backend trusts a client-supplied resource id without verifying ownership.
- **Audit logging** provides traceability and is the first tool for incident response.